

Docker Installation

Install Docker Desktop, then use Docker from the command line.

Learning Goals

- Choose the right terminal for your computer (WSL2 on Windows, Terminal on macOS).
- Check the Windows and WSL2 requirements before installing Docker Desktop.
- Install Docker Desktop and verify it with `docker version` and `docker compose version`.

Choose Your Terminal

Computer	Install from	Run course commands in
Windows	PowerShell	Your WSL2 distribution
macOS	Terminal	Terminal

- Window users: This lesson assumes you already use WSL on Windows.

For Windows Users: Check Before Installing

Docker for Windows uses WSL2

- You need a supported Windows version (Windows 10 22H2 or Windows 11 23H2 or later), 8 GB RAM, and hardware virtualization enabled in BIOS/UEFI.

Check WSL version in PowerShell

```
wsl --version
```

```
PS C:\Users\smcho> wsl --version  
WSL version: 2.7.12.0  
Kernel version: 6.18.33.2-2  
1-240331-1435.ge-release  
Windows version: 10.0.26200.9168
```

```
wsl --list --verbose
```

```
PS C:\Users\smcho> wsl --list --verbose
```

NAME	STATE	VERSION
* Ubuntu-24.04	Running	2

- WSL itself must be **2.1.5 or later**; your Linux distribution must show **VERSION 2**.
- If version details do not appear or WSL is older, run `wsl --update` .
- If the distribution shows VERSION 1, run `wsl --set-version <NAME> 2` with its exact NAME from the list (in this example, Ubuntu-24.04).

Docker Desktop

1. Download **Docker Desktop Installer** from the <https://docs.docker.com/desktop/>
2. Install the Docker Desktop for your OS

Sign up (if necessary) and sign in to use the Docker Desktop.

macOS: Install with Homebrew

If [Homebrew](#) is already installed, use Terminal:

```
brew install --cask docker-desktop  
open -a Docker
```

`brew install docker` installs the CLI alone and does not provide a local engine. The `docker-desktop` cask requires macOS 14 or later.

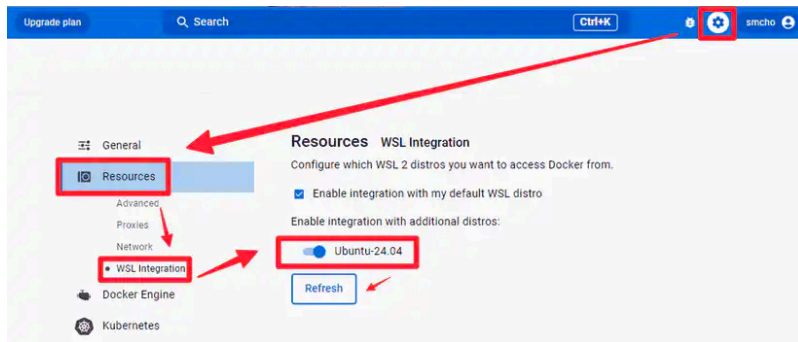
macOS: docker path

The Docker CLI is installed in `~/ .docker/bin .`

- Add this path to your `PATH` environment variable to use the Docker command-line interface.

```
> which docker  
/Users/chos5/.docker/bin/docker
```

Windows: WSL2 Integration



Open **Settings** → **Resources** → **WSL Integration**. Enable the distribution you use, then select **Apply & restart**.

Windows: Use Docker from PowerShell or WSL2

Docker Desktop uses WSL2 behind the scenes.

- After enabling WSL Integration for your distribution, you can run Docker commands from either PowerShell or that WSL2 shell.

Windows: Adding user to the docker group

You need to add current user to the docker group to run docker commands on WSL2.

- This step can still be required even when using Docker Desktop, if you see a permission error accessing the Docker socket.

```
permission denied while trying to connect to the docker API at  
unix:///var/run/docker.sock
```

```
sudo usermod -aG docker $USER  
newgrp docker
```

Check the Result

```
docker version
```

```
Client:
  Version:           29.8.1
  API version:       1.56
  Go version:        go1.26.8
  Git commit:        4a63305
  OS/Arch:           darwin/arm64
  Context:           desktop-linux

Server: Docker Desktop 4.93.0 (240920)
Engine:
  Version:           29.8.1
  API version:       1.56 (minimum version 1.40)
  Go version:        go1.26.8
  OS/Arch:           linux/arm64
  Experimental:      false
```

WSL2 example:

Client:

Cloud integration: v1.0.29
Version: 20.10.22
API version: 1.41
Go version: go1.18.9
OS/Arch: windows/amd64
Context: default

Server: Docker Desktop 4.16.3 (96739)

Engine:

Version: 20.10.22
API version: 1.41 (minimum version 1.12)
Go version: go1.18.9
Git commit: 42c8b31
Built: Thu Dec 15 22:26:14 2022
OS/Arch: linux/amd64
Experimental: false

Updating the Docker

If the current version is too old or needs update, you can use command line:

```
# Mac  
brew upgrade --cask docker-desktop  
  
# Windows  
winget upgrade Docker.DockerDesktop
```

Or, use Docker Desktop software update:

Settings → Software Updates

Check docker compose

```
> docker compose version  
Docker Compose version v5.5.1
```

- Docker manages containers and images.
- Docker Compose defines and runs multi-container applications.
- Use `docker compose` with a space; `docker-compose` with a hyphen is the legacy command.

First docker example

```
docker run --rm hello-world
```

```
latest: Pulling from library/hello-world  
58dee6a49ef1: Pull complete  
c3bdf82c34d1: Download complete  
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8  
Status: Downloaded newer image for hello-world:latest
```

```
Hello from Docker!  
This message shows that your installation appears to be working correctly.
```

The `Hello from Docker!` message is from the hello-world Docker image.

`docker run` creates and starts a container from the `hello-world` image.

- If the image is not available locally, Docker downloads it first. The `--rm` option automatically removes the container after it finishes.
- Use the Docker desktop to check the docker image is downloaded and removed from the container with the `--rm` option.

